The basement membrane as a surface with a reservoir:

deformation, remeshing, tearing, and the Plexus2 container they live in

`prototype/ecm/bm_ops.py` · `test_05_sheet.py` · `test_05b_plaque.py` · `test_05c_remesh.py` · `test_05e_conserve.py` · `log/okuda_ECM/05a-05e` · 10 August 2026

1. Mechanism

A basement membrane enclosing a growing spheroid has to do two things that pull against each other: *hold together as a material*, and *keep covering a surface whose area goes up by an order of magnitude.* Sixty-six archived runs modelled it as MPM material and could do neither, for two reasons that turn out to be the same reason.

It could not report its own deformation. In MPM the deformation gradient $\mathbf{F}$ is advected by the *grid's* velocity gradient, so a node moved by anything that is not the grid — an engine delta, a positional projection — carries no record of having moved. Run 130 moved the sheet out with the epithelium by a factor of 3.44 in radius, a strain of 2.44, and its material reported 0.31: thirteen per cent. Run 121 routed the same force through the grid, recovered 2.25 of 2.30, and paid for it with the standoff. A sheet whose strain is wrong by a factor of eight cannot be given a stiffness, cannot tear at a stress, cannot thin by an area ratio and cannot be told whether its adhesion is load-bearing — every downstream question is unanswerable rather than merely inaccurate.

It could not change size. The number of MPM particles was fixed, so `basement_membrane_secrete` added material that was never part of the sheet: run 128's 45,000 new particles sat at standoff -0.19, a second shell in the lumen, and halved the reported strain.

Both are consequences of the sheet not being an *object with a topology.* This note builds it as one: a codimension-1 mesh whose deformation is measured rather than advected (§2–§3), whose material memory and supply are explicit (§4), whose adhesion is a relation rather than a body (§5), and whose *size is a state variable* — a reservoir of triangles with an occupancy flag (§6). §7 writes all of it as a Plexus2 container; §8 is the gate table; §9–§10 are what the runs measured, including three gates that fail.

One distinction is load-bearing throughout, and an earlier draft of this note blurred it. Refinement (`bm_refine`) is a *numerical* operation: it changes the triangulation and nothing else — not the physical surface, not its material state, not its mass. Secretion, turnover and proteolysis (`bm_secrete`, `bm_remodel`, `bm_degrade`) are *biological* mechanisms that change what the sheet is made of and how much of it there is. They share an implementation substrate, the `occ` reservoir, and they share nothing else. Writing them as "the same mechanism seen three times" — which the first draft did — is exactly the confusion that lets a discretisation choice be read as a result, and it is the same error as the MPM sheet whose tearing was set by Δx. Everything a remesh must carry across unchanged is listed in §6 and gated in §8.

The second reason to leave MPM is arithmetic and is in `note_fibre` §9: a basement membrane is $\sim$100 nm on a $\sim$200 μm spheroid, one part in two thousand of the radius, so resolving it on a grid needs $\Delta x \approx 3 \times 10^{-4}$ and a lattice of 3500^3. It is not a thin solid we cannot afford. It is a *surface*, and its thickness belongs in the constitutive law as a number, never in the geometry as a length.

2. Equations and symbols

λ is the quantity every other statement here is written in, and it is a *per-triangle* number. For a face with nodes $\mathbf{x}_{0,1,2}$, take the two edge vectors from the first node now and in the reference configuration $\mathbf{X}$:

$$\mathbf{D}_s = \left[\, \mathbf{x}_1 - \mathbf{x}_0,\ \mathbf{x}_2 - \mathbf{x}_0 \,\right] \in \mathbb{R}^{3\times 2}, \qquad \mathbf{D}_m = \begin{bmatrix} a & \mathbf{v}\cdot\hat{\mathbf{e}}_1 \\ 0 & \mathbf{v}\cdot\hat{\mathbf{e}}_2 \end{bmatrix} \in \mathbb{R}^{2\times 2}, \tag{1}$$

where $\mathbf{D}_m$ is the same pair written in the reference triangle's *own* in-plane frame $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2)$, with $a = |\mathbf{X}_1 - \mathbf{X}_0|$ and $\mathbf{v} = \mathbf{X}_2 - \mathbf{X}_0$. Then

$$\mathbf{F} = \mathbf{D}_s \mathbf{D}_m^{-1} \in \mathbb{R}^{3\times 2}, \qquad \mathbf{C} = \mathbf{F}^\top \mathbf{F}, \qquad \lambda_{1,2} = \sqrt{\operatorname{eig} \mathbf{C}}, \qquad J = \lambda_1 \lambda_2 = \frac{\text{area now}}{\text{area then}}. \tag{2}$$

λ_1 is the larger principal stretch: 1 is unstretched. **This is what the movies colour by** — `lam` in `traj.npz`, `lam_geo` in `metrics.json`. Because $\mathbf{F}$ is read from where the nodes *are* against where they *were*, it is blind to what moved them: a node moved by a tether, a plaque, a grid or by hand is seen identically, and there is no path by which a displacement is applied and not counted. That is a property of Eq. (2) and not a result, which is why it can be certified in closed form (§8, G1–G4) before any physics runs.

There are two λ's and they must never be quoted as one. The reference metric is either frozen at seeding or allowed to creep (§4), giving

$$\lambda^{\rm geo} = \sqrt{\operatorname{eig}\mathbf{C}}\ \text{(against the SEEDED metric)}, \qquad \lambda^{\rm el} = \sqrt{\operatorname{eig}\big(\mathbf{C}_0^{-1}\mathbf{C}\big)}\ \text{(against the REMEMBERED metric } \mathbf{C}_0\text{)}. \tag{3}$$

$\lambda^{\rm geo}$ is what the tissue did to the sheet; $\lambda^{\rm el}$ is what the material feels. With no remodelling they coincide. With remodelling they do not, and the gap is the mechanism.

2.1 The mechanics

One constant-strain triangle per face, St Venant–Kirchhoff, thickness entering only as a multiplier:

$$W_f = A_f^0\Big[\mu\,\mathrm{tr}\big(\mathbf{E}_g^2\big) + \tfrac{\Lambda}{2}\big(\mathrm{tr}\,\mathbf{E}_g\big)^2\Big], \qquad \mathbf{E}_g = \tfrac{1}{2}\big(\mathbf{C}_{\rm el} - \mathbf{I}\big), \qquad \mathbf{f} = -\frac{\partial}{\partial\mathbf{x}}\sum_f W_f, \tag{4}$$

$$Y_2 = ET \quad \text{[force per unit } \textit{length}\text{]}, \qquad \mu = \frac{Y_2}{2(1+\nu)}, \qquad \Lambda = \frac{Y_2\,\nu}{1-\nu^2}. \tag{5}$$

The force is the exact gradient, taken by automatic differentiation, so the force and the energy cannot disagree — G4 checks it against a central finite difference and gets 5×10^{-10} relative.

Massless and overdamped, so the stability limit is a rate and not a wave speed:

$$\gamma\dot{\mathbf{x}} = \mathbf{f}, \qquad \mathbf{x} \leftarrow \mathbf{x} + \Delta t\, M\mathbf{f}, \quad M = 1/\gamma, \qquad \text{stable iff } \boxed{\Delta t\, M\, \lambda_{\max}(\mathbf{H}) < 2} \tag{6}$$

with $\mathbf{H}$ the Hessian of Eq. (4). This is about twenty times weaker than the CFL an inertial sheet would carry, and it is why a sheet ten to a hundred times stiffer than the stroma does not set the timestep for the whole model. $\lambda_{\max}$ is *measured*, by power iteration on Hessian-vector products, and re-measured every ten frames, because it is not a constant: a StVK membrane's tangent stiffens as it stretches, and over 401 frames it grew 19.1× (4.56 → 87.2), taking the substep count with it (21 → 194). The first version of the rig fixed the substep count from the seeded value and put the divergence threshold at 1.8 instead of 2 — the entire discrepancy being the Hessian having grown under the run's own load. Held at a constant group $s = \Delta t M \lambda_{\max}$ instead, the sweep runs to 1.9 and diverges at 2.1.

2.2 Memory, stiffening, supply

Remodelling is the reference metric creeping toward the current one, the 2D form of the archived per-edge $\dot{\ell}^0 = (L - \ell^0)/\tau_r$:

$$\mathbf{C}_0 \leftarrow \mathbf{C}_0 + \frac{\Delta t}{\tau_r}\big(\mathbf{C} - \mathbf{C}_0\big). \tag{7}$$

Collagen IV's half-life of 3–10 hours is 13–93 frames here, so this is resolved rather than instantaneous. A sheet enclosing a surface that triples in radius cannot do it elastically.

Strain stiffening, $E(\lambda) = E_0[1 + \beta(\lambda^{\rm el} - 1)]$, applied explicitly from the previous step's stretch so that Eq. (4) remains the exact potential of the force. Candiello *et al.* measure 0.4 → 3 MPa on native basement membrane; **MPa is not a quantity this box has** — it is dimensionless and the stroma's modulus is the number 15 — so the falsifiable forms are the ratio $E(\lambda_{\max})/E_0$ and the contrast $E_{\rm bm}/E_{\rm ecm} \in [10, 100]$.

Areal density, and which of the two is the state. The mass balance a growing sheet owes is usually written for ρ,

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} = \underbrace{s(\mathbf{x},t)}_{\text{secretion}} - \underbrace{\frac{\rho}{\tau_{\rm bm}}}_{\text{turnover}} - \underbrace{\rho\frac{\dot{A}}{A}}_{\text{dilution}}, \qquad s^* = \rho\Big(\frac{1}{\tau_{\rm bm}} + \frac{\dot{A}}{A}\Big), \tag{8}$$

whose two removal terms are the same size here ($\sim 10^{-1}\,\mathrm{h}^{-1}$ each) — turnover is not a correction to growth, it is the same size as growth, which is why a sheet can accommodate growth at all.

But ρ is the wrong variable to store, and 05e is the run that changed it. Carry the *mass* m_f on each face and derive the density,

$$\rho_f = \frac{m_f}{A_f(\text{now})}, \qquad \frac{\mathrm{d}m_f}{\mathrm{d}t} = s\,A_f - \frac{m_f}{\tau_{\rm bm}}, \tag{9}$$

and three things follow that the ρ form does not give. The dilution term *disappears from the integration* — it is what dividing by a growing area does on its own, and Eq. (8) integrated alongside Eq. (9) would count it twice.

$\sum_f m_f$ becomes a conserved quantity that only secretion and proteolysis may change, so a remesh can be *gated* on it (G18b, measured bit-identical). And run 128's failure mode becomes structurally impossible: mass is added to faces that already are the sheet, so there is no way to secrete material into the lumen.

With $s = 0$ the measured sheet thins to 1/11.1 of what it was seeded with, which is Eq. (9) with no source and an area $\times 11.1$. **That number is the baseline `bm_secrete` has to hold flat**, and it is the reason a fixed particle budget was never going to work.

2.3 The plaque, which is a relation and not a body

An integrin plaque has no position of its own; it is an edge between two entities, and writing it as an edge set is what makes its reaction automatic — one operator returns a delta to both endpoints, so the force the sheet feels and the force the cell feels cannot get out of step. Each plaque binds a membrane node to a *barycentric point on an epithelial face*, $\mathbf{p} = \sum_k w_k \mathbf{x}_k^{\text{epi}}$, which tracks the material as that face deforms:

$$\boldsymbol{\ell} = \mathbf{x}^{\text{bm}} - \mathbf{p}, \quad \ell = |\boldsymbol{\ell}|, \quad \hat{\mathbf{n}} = \boldsymbol{\ell}/\ell, \qquad \mathbf{f}_n = -\kappa_n(\ell - \ell_0)\hat{\mathbf{n}}, \tag{10}$$

$$\mathbf{f}_t = -\xi\big(\mathbf{v}_{\text{bm}} - \mathbf{v}_{\text{epi}}\big)_{\parallel}, \qquad \mathbf{f}_k^{\text{epi}} = -w_k\big(\mathbf{f}_n + \mathbf{f}_t\big), \tag{11}$$

at the measured density Σ^{-2} with $\Sigma \approx 7T$ and $\ell_0 \approx 0.3\,T$ [3].

The tangential law is solved, not evaluated. An explicit $-\xi\mathbf{v}$ is a stiffness of $\xi/\Delta t$, which *grows* as the substep shrinks, so refining the step to stabilise it makes it worse — the first version NaN'd on frame 0. Solving $\dot{\mathbf{x}}_\parallel = M(\mathbf{f}_\parallel + \xi\mathbf{v}_\parallel^{\text{epi}})$ for $\dot{\mathbf{x}}_\parallel$ instead gives

$$\dot{\mathbf{x}}_\parallel = \frac{\dot{\mathbf{x}}_\parallel^{\text{prov}} + M\xi\,\mathbf{v}_\parallel^{\text{epi}}}{1 + M\xi}, \tag{12}$$

unconditionally stable in ξ and tending to no-slip as $\xi \to \infty$; the force actually applied is recovered from the solved velocity and handed to both bodies, so this costs nothing in the momentum gate.

2.4 The reservoir: a sheet whose size is a state variable

This is the part the archived line had no equivalent of, and it is what makes the rest possible.

What a remesh is, and what it must never be. `bm_refine` is a numerical topology update. The physical surface is unchanged — the same points of space, the same enclosed volume, the same material — and so is everything the material knows about itself. It exists because a fixed triangulation stops resolving a surface whose area grows by an order of magnitude, and for no other reason. **Nothing biological may be inferred from a refinement, and no biological operator may be implemented as one.** A tear that appears because the mesh was coarse, a thinning that appears because triangles grew, a stiffening that appears because elements changed shape — each of those is a discretisation artefact wearing a mechanism's name, and the MPM sheet whose separation was set by Δx is the cautionary case.

What must be carried across a remesh, unchanged or conservatively. Three things, and each is gated:

carried	how	why it is load-bearing
$\mathbf{C}_0$, the remodelling state	inherited identically ($\mathbf{C}_0$ is intensive and lives in material coordinates)	it is the sheet's memory of what it has forgotten; resetting it makes a refined sheet elastically virgin
m_f, the mass	extensive: each child takes a quarter, so $\sum_f m_f$ is invariant (**measured bit-identical**)	$\rho = m_f/A_f$ is then derived, so a split that does not conserve mass would be a secretion nobody asked for
plaque attachments	node indices survive (nodes are appended, never removed); the plaques are RE-SEEDED to hold Σ^{-2}	**measured**: held at 0.996 of Σ^{-2}, against 0.145 for the fixed-fraction rule 05b used

Both of the last two rows were defects when this section was first written, and 05e is the run that fixed them. The plaque one is worth a number: 05b's rule — anchor a fixed fraction of the nodes — did not lose a factor of 4 per refinement as predicted, it lost **a factor of 6.9 over the run**, because the fraction is wrong twice over: once at each split, and continuously as the area grows between them. `plaque_seed` now takes Σ in units of T and tops the count up to $A(t)/\Sigma^2$ every frame, which is also the biology — a growing epithelium makes more hemidesmosomes, it does not stretch the ones it has. Every plaque number in `05b_plaque` is therefore a number at the wrong density, and its density sweep reads as three arbitrary densities rather than as 100% / 20% / the measured one.

The problem, as a number. With a fixed mesh the sheet's area goes $\times 11.1$ over 401 frames on the same 5120 triangles: the mean edge length goes $0.00305 \to 0.0102$ box units, $3.4\times$ its seeded value, and each triangle covers eleven times the tissue it started on. A mass balance cannot add material to a set whose size is fixed, and a proteolytic operator cannot remove it.

The pattern already exists in the framework. `engine.py:453` allocates `grow_reserve` dormant MPM particles at `occ` $= 0$ for `cell_grow` to wake, and `engine.py:134` counts a set's live members as (`occ` > 0). The sheet now carries the same flag on *both* of its sets. Nodes and faces are allocated to their maximum at seeding — 40,962 node slots and 81,920 face slots for a sheet that starts at 2562 and 5120 — and every operator that changes the sheet's size flips `occ` rather than reallocating. Three consequences, and the third is the reason to do it this way rather than by growing arrays:

1. `bm_refine` wakes three faces per parent and one node per edge;
2. `bm_degrade` puts faces to sleep, and a hole is then simply a region with no live faces;
3. **an index survives all of it.** A plaque holds a *node* index, and nodes are only ever appended and never removed, so no adhesion is invalidated by a refinement or by a tear. Had the plaque bound to a face, every one of them would need reconciling after every split.

Refinement is a global $1\to 4$ midpoint split of every live face, triggered when the mean live edge length passes a multiple of its seeded value. It is conforming by construction — every edge is split, so no face is left with a hanging node on one of its sides — and needs no red–green closure. It is a *material* trigger, not a schedule: a sheet that is not being stretched never refines.

The trap, and the identity that avoids it. A child triangle whose reference frame is rebuilt from its *current* shape reports $\lambda = 1$ the moment it is born, so a sheet that refines as it grows would silently forget everything it had been stretched by — the mesh version of the laundering that made run 130 report 13% of its true stretch. The four children of a midpoint split are related to their parent by a *constant* map $\mathbf{S}_k$ in material coordinates, so

$$\mathbf{D}_m^{\text{child}} = \mathbf{D}_m^{\text{parent}}\mathbf{S}_k \implies \mathbf{D}_m^{-1,\text{child}} = \mathbf{S}_k^{-1}\mathbf{D}_m^{-1,\text{parent}}, \qquad A^0 \to \frac{A^0}{4}, \quad \mathbf{C}_0,\ \rho,\ Y_2 \text{ inherited,} \tag{13}$$

with, for children $(\mathbf{v}_0, \mathbf{m}_{01}, \mathbf{m}_{20})$, $(\mathbf{v}_1, \mathbf{m}_{12}, \mathbf{m}_{01})$, $(\mathbf{v}_2, \mathbf{m}_{20}, \mathbf{m}_{12})$, $(\mathbf{m}_{01}, \mathbf{m}_{12}, \mathbf{m}_{20})$,

$$\mathbf{S}_{1..4} = \tfrac{1}{2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix},\ \tfrac{1}{2}\begin{bmatrix}-1 & -1\\ 1 & 0\end{bmatrix},\ \tfrac{1}{2}\begin{bmatrix}0 & 1\\ -1 & -1\end{bmatrix},\ \tfrac{1}{2}\begin{bmatrix}0 & -1\\ 1 & 1\end{bmatrix}, \qquad \det\mathbf{S}_k = \tfrac{1}{4} > 0\ \ \forall k. \tag{14}$$

Every $\det\mathbf{S}_k$ is positive, so orientation is inherited too. Substituting Eq. (13) into Eq. (2) gives $\mathbf{F}^{\text{child}} = \mathbf{F}^{\text{parent}}$ *identically*, and that identity is gate G15: measured at 2.3×10^{-14} on a loaded sheet, with the total area and the total elastic energy **bit-identical** across the split. Refinement is not approximately invisible to the mechanics; it is exactly invisible.

One deliberate omission. The midpoints are placed on the *chord* and are not projected onto the surface. Projecting would smooth the sheet and change λ, breaking Eq. (13) by construction: the smoothing has to come from the dynamics, not from the remesher.

Tearing is the same mechanism with the flag going the other way. A face dies when a criterion is met — λ^{el} past a break strain, the second Piola stress past a yield, or (the version the biology prefers) ρ below a critical areal density, which couples the tear to Eq. (8): a basement membrane fails where it is not resupplied. The rim of the hole is free, no node is removed, and the mesh stays conforming: opening a polar cap of 106 faces produced 34 rim edges and zero edges carrying three faces.

Why this needs a gate that MPM's tearing never had. `LADDER.md:24` flagged it at the outset: "MPM separates material that thins past the grid's support, so tearing may be set by dx rather than by stress. A tear that moves when the grid is refined is numerical." The refinement cross-check (run 90, n_{grid} $64 \to 128$) was planned and never ran, on a sheet 1/8 of a grid cell thick that was carrying the stroma's `youngs` 15 rather than the 400 its spec declared. So the visually convincing MPM tearing is, as of today, untested; G19 in §8 is that test.

2.5 Symbols

Geometry and strain — Eqs. (1)–(3)		
$\mathbf{x}_i$	node position	one corner of a triangle; the state of set `bm`
$\mathbf{D}_s, \mathbf{D}_m$	current, reference edge matrices	of one face; $\mathbf{D}_m$ is in that face's own 2D frame
$\mathbf{F}$	deformation gradient	3×2: maps the reference triangle to the current one
$\mathbf{C}, \mathbf{C}_0$	current, remembered metric	$\mathbf{C} = \mathbf{F}^\top \mathbf{F}$; $\mathbf{C}_0$ starts at $\mathbf{I}$ and creeps
λ_1, λ_2	principal stretches	$1 =$ unstretched. *Per face*, never per node
$\lambda^{\text{geo}}, \lambda^{\text{el}}$	geometric, elastic stretch	what the tissue did / what the material feels
J	areal ratio	$\lambda_1 \lambda_2$
A_f^0	reference area	of face f
Material — Eqs. (4)–(5), (7)–(8)		
E	Young's modulus	force per unit *area*; meaningless here without T
T	thickness	enters ONLY as a multiplier; never resolved as geometry
$Y_2 = ET$	2D stretch modulus	force per unit *length* — the sheet's actual stiffness
ν; μ, Λ	Poisson ratio; 2D Lamé	in the plane of the sheet
β	stiffening coefficient	$E(\lambda) = E_0[1 + \beta(\lambda - 1)]$
τ_r	remodelling time	how fast the sheet forgets being stretched, IN FRAMES
$\rho,\ \tau_{\text{bm}},\ s$	areal density, turnover time, secretion rate	of collagen IV, Eq. (8)
Integration — Eq. (6)		
$M = 1/\gamma$	mobility	force to velocity; the sheet has no mass, so only M appears
$\lambda_{\max}(\mathbf{H})$	largest Hessian eigenvalue	MEASURED, every ten frames; it grew 19.1× over a run
$s = \Delta t M \lambda_{\max}$	stability group	the bound is $s < 2$; the substep count is set to hold it
ζ	relaxation e-folds per frame	does the sheet relax faster than the tissue grows
Reservoir — Eqs. (13)–(14)		
`occ`	occupancy flag	$1 =$ live, $0 =$ dormant. On `bm` AND on `bm_face`
$\mathbf{S}_k$	child map	constant 2×2 in material coordinates; $\det = \frac{1}{4}$
$\mathbf{m}_{ij}$	midpoint node	of the edge between nodes i and j, placed on the CHORD
Plaque — Eqs. (10)–(12)		
ℓ, ℓ_0	current, rest length	of one plaque; $\ell_0 = 0.3\,T$
κ_n	**normal stiffness**	of the plaque (not the sheet's Y_2, not the tether's κ)
ξ	tangential friction	against RELATIVE velocity of sheet and epithelium
w_k	barycentric weights	of the attachment point in its epithelial face; also the reaction's weights
Σ	plaque spacing	$\approx 7T$ measured; the density is Σ^{-2}
κ	tether stiffness	`05a` only: the one-sided spring the plaque replaces

3. The Plexus2 container

Two questions decide whether any of the above can be written down: what the *sets* are, and what kind of operator each mechanism is.

Sets, and one structural claim. A triangle is not a body: it is a *relation among three nodes*, exactly as a junction is a relation between two vertices in the epithelium. So the sheet is two sets, not one — `bm` carries position and `bm_face` carries everything material — and `bm_face` is a hyperedge set of arity 3 with `.a`, `.b`, `.c` index buffers, the same shape as `junction` $\subseteq$ `vertex`^{2} and `plaque` $\subseteq$ `cell`$\times$`bm`. Putting the material state on the faces rather than on the nodes is what makes Eq. (13) a local inheritance rule instead of an interpolation.

set	kind	one element is	state it carries
`sheet`	node, depth 0	the membrane as an organ	total area, $\langle\rho\rangle$, live counts
`bm`	node, depth 1	one node of the membrane	`pos`, `vel`, `occ`
`bm_face`	**hyperedge** $\subseteq$ `bm`^{3}	one patch of membrane	$\mathbf{D}_m^{-1}$, $\mathbf{C}_0$, A^0, ρ, Y_2, age, `occ`
`plaque`	**edge** `cell`$\to$`bm`	one hemidesmosome	ℓ_0, κ_n, ξ, barycentric w_k, load, bound
`fibril`	**edge** `bm`$\to$`ecm`	one anchoring fibril	rest length, load
`protease`	field	MMP concentration	scalar, diffusing (drives `bm_degrade`)

Operators, in Plexus2's taxonomy: *Seed* creates a set, *Lateral* moves state along an edge set, *Exchange* scatters to or gathers from a field, *Rewire* changes which elements are connected, and *Divide*/*Die* change how many exist. The reservoir is what lets the last two be $O(1)$ flips.

operator	kind	acts on	mechanism	its gate
`bm_seed`	Seed	`bm`, `bm_face`	lay the sheet on the recorded surface; allocate the reservoir	frame 0 reports $\lambda = 1$
`bm_elastic`	Lateral	`bm_face`	Eqs. (4)–(5), force as the exact gradient	G1–G5
`bm_stiffness`	Lateral	`bm_face`	$E(\lambda) = E_0[1 + \beta(\lambda - 1)]$	E/E_0 ratio, not MPa
`bm_remodel`	Lateral	`bm_face`	Eq. (7)	$\lambda^{\mathrm{el}} \to 1$ while λ^{geo} holds
`bm_refine`	**Divide**	`bm_face`	*numerical maintenance only*: Eqs. (13)–(14), wake 3 faces + 1 node per edge	G14–G18c
`bm_secrete`	Lateral	`bm_face`	Eq. (9): add MASS to existing faces [4]	G24–G26
`bm_degrade`	**Die**	`bm_face`	proteolysis, or ρ below critical [5]	G19–G23
`plaque_seed`	Rewire	`plaque`	density Σ^{-2}, $\Sigma \approx 7T$ [3]; turnover re-seeds	plaque spacing in T
`plaque_pull`	Lateral	`plaque`	Eqs. (10)–(12), *with its reaction*	G9–G13
`plaque_rupture`	Rewire	`plaque`	the bond fails past a load	fraction bound vs. load
`fibril_pull`	Lateral	`fibril`	the stroma is loaded *through* the sheet [7]	stroma displacement, sheet present vs. absent
`bm_barrier`	Exchange	`bm_face`, fields	pores pass solutes, stop cells [6]	flux across an intact vs. breached sheet

Schedule, and the division is not cosmetic. Anything that carries a force runs *inside* the substep block, at the sheet's own rate; anything that changes the sheet's topology or its mass runs once per frame, because a set whose membership changed mid-substep would be integrated against two different index sets in one step:

```
schedule:
  - bm_refine           # frame level: topology. Trigger = mean live edge > 1.45 x seeded
  - bm_secrete          # frame level: mass.     Eq. (10), wakes faces
  - bm_degrade          # frame level: mass.     Eq. (10) / stress, sleeps faces
  - plaque_seed         # frame level: rewire.   turnover, which is what permits slip
  - {substep_dt: 4.0e-4, steps: [bm_stiffness, bm_elastic, plaque_pull, fibril_pull,
                                 mpm_strain, mpm_scatter, mpm_grid_update, mpm_gather]}
  - bm_remodel          # frame level: Eq. (9), tau_r in frames
  - plaque_rupture      # frame level: rewire on the load the substeps just produced
```

The substep count is not a constant: it is set from the measured $\lambda_{\max}$ to hold $\Delta t M \lambda_{\max}$ at a chosen group below 2 (§3), which over a run means $21 \to 194$.

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. G1–G4 and G14–G16 run before any physics, as `bm_ops.selftest()`; a rig refuses to start if they fail.

	gate	threshold	measured
the strain measure (§2)			
G1	rigid motion leaves $\lambda = 1$	$< 10^{-6}$	7.5×10^{-9}
G2	dilation by s: $\lambda_1 = \lambda_2 = s$, every face	$< 10^{-6}$	2.5×10^{-8} (J: 1.1×10^{-13})
G3	affine map gives the singular values of $\mathbf{A}$	$< 10^{-10}$	8.9×10^{-15}
G4	force $= -\partial W/\partial \mathbf{x}$, vs finite difference	$< 10^{-6}$ rel.	5×10^{-10}
the sheet under load (`05a_sheet`)			
G5	stretch fidelity vs the applied $R(t)/R(0)$	> 0.95	0.9935 (MPM: 0.13, 0.98)
G6	control $\kappa = 0$: nothing pulls, nothing stretches	$= 1.00000$	1.00000
G7	control, rigid drive: motion is not deformation	$= 1.00000$	1.00000
G8	divergence at $\Delta t M \lambda_{\max} = 2$	stable < 2, NaN > 2	**ran at** 1.9**, NaN at** 2.1
the plaque (`05b_plaque`)			
G9	momentum, bookkeeping: $\lvert\sum \mathbf{f}_{\mathrm{bm}} + \sum \mathbf{f}_{\mathrm{epi}}\rvert / \sum \lvert\mathbf{f}\rvert$	$< 10\,\varepsilon_{64}$	1.4×10^{-16}
G10	momentum, motion: free pair, centroid drift	$< 10^{-12}$	4.4×10^{-15}
G11	control, no plaques: the sheet is not carried	$\lambda \approx 1$	1.25 vs 3.30 (marginal)
G12	standoff equals ℓ_0	$\ell/\ell_0 \in [0.9, 1.1]$	3.43 (§10)
G13	slip: $\xi = 0$ and $\xi > 0$ must differ	any separation	**none** (§10)
the reservoir (§6, `05c_remesh`, `05e_conserve`)			
G14	refining an UNLOADED sheet changes λ	$< 10^{-6}$ $(\sqrt{\varepsilon})$	6.2×10^{-9}
G15	refining a LOADED sheet: λ / area / energy	10^{-12}, 10^{-9}, 10^{-6}	2.3×10^{-14}**,** 0**,** 0
G16	no hanging nodes	exact	0 **bad,** 0 **rim**
G17	mean edge stays in a band as R triples	$[0.8, 1.7]\times$ seeded	0.84 (fixed mesh: 3.33)
G18	refinement does not degrade fidelity	$\geq$ the fixed mesh's	0.9985 **vs** 0.9873
G18b	mass $\sum_f m_f$ invariant across a split	exact	0**, bit-identical**
G18c	plaque areal density held at Σ^{-2}	within 10%	0.996 vs control 0.145
G18d	a split does not move the surface: volume, area, centroid	$< 10^{-12}$ rel.	2.6×10^{-16}**,** 0**,** 9×10^{-17}
G18e	$\rho = m/A$ equals $1/J$	—	an identity, not a test
tearing (`05d_tear`) — G19 is the test the MPM sheet never faced			
G19	**same tear, same λ, under refinement**	onset λ within 5%	0.52% (2.110 vs 2.099)
G20	onset scales with the CRITERION, not element size	monotone / flat	**frame** 167/227/285 **at** 1.8/2.2/2.6
G21	a hole stays open; the remesher does not heal it	exact	**structural** (rim persists to the last frame)
G22	control: criterion above the load kills nothing	exact	0 **of** 81,920
G23	the mesh is conforming with a hole in it	0 edges with 3 faces	0
supply (§4, `05f_secrete`)			
G24	ρ held flat while A triples	within 10%	05f, running
G25	the added material is in the sheet; polarised follows the plaques	by construction; $r > 0$	05f, running
G26	starved tears, fed does not, at the same λ	—	05f, running
G27	dilution and turnover are comparable	ratio within $[0.3, 3]$	05f, running

Two conventions make these mean what they say. **The momentum residual is taken on the adhesion's own force pair**, inside the substep, before the drive is added — with the drive's force in the sum it stops being a statement about the reaction. And **a rupture threshold is taken from the load distribution the nominal run measured**, never from a round number: run 127's `detach: 0.02` sat above the largest load in its entire run (0.017), so a correctly wired mechanism was reported as a null.

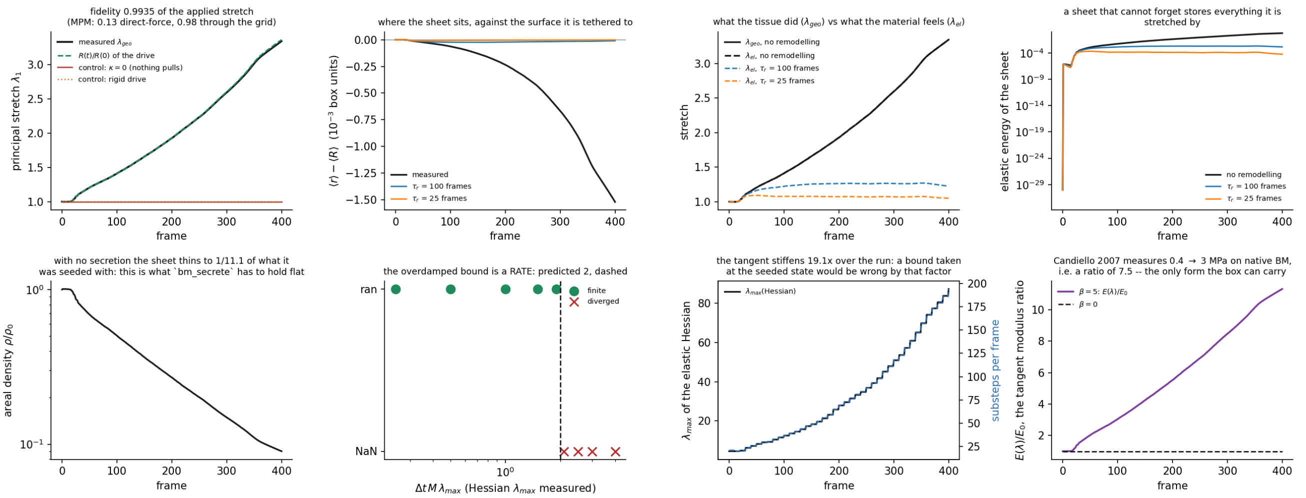

Figure 1: `05a_sheet`, the sheet alone on the recorded epithelial surface. **a G5**: the measured λ^{geo} lies on the applied $R(t)/R(0)$ to 0.9935, against 0.13 for the MPM sheet moved by an engine delta; the two flat lines at 1.0000 are **G6** ($\kappa = 0$) and **G7** (a rigid drive), i.e. the measure invents no strain and does not mistake motion for deformation. **b–c** remodelling: at $\tau_r = 25$ frames the standoff improves 1026× while λ^{geo} is unmoved — the sheet still goes where the tissue puts it and stops fighting. **f G8**: divergence at the predicted group $\Delta t M \lambda_{\max} = 2$, measured, not assumed. **g** the tangent stiffens 19× over the run, which is why the bound has to be re-measured. **h** the stiffening law as a ratio, since MPa is not a quantity this box has.

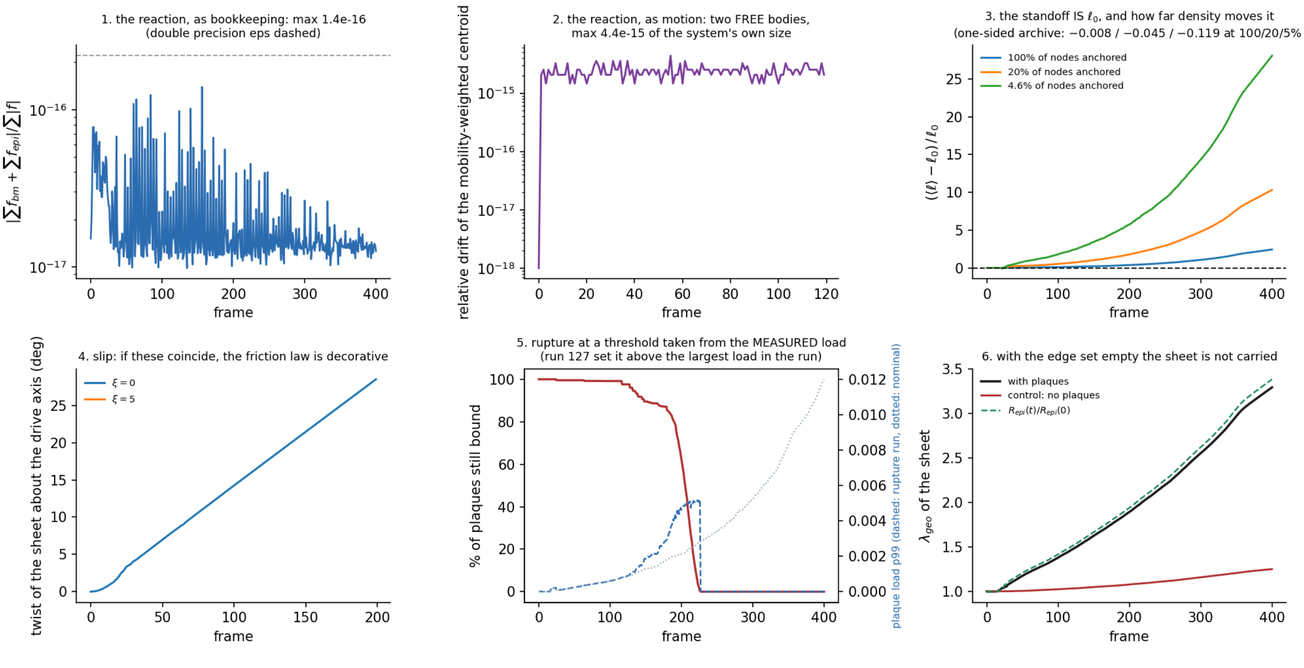

Figure 2: `05b_plaque`, the adhesion as an edge set. **1 G9**: the residual on the adhesion's own force pair sits at 1.4×10^{-16}, i.e. at double-precision ε — one operator, a delta to both endpoints, and the reaction cannot be half-implemented. **2 G10**: the same claim as motion, with two free bodies whose mobility-weighted centroid drifts by 4.4×10^{-15} of the system's own size. **3 G12**: the standoff is $3.43\,\ell_0$ and not ℓ_0. **4 G13**: $\xi = 0$ and $\xi > 0$ do not separate, because a fixed barycentric anchor is a tangential pin. **5** rupture is a clock rather than a load, for want of turnover. §10 is what each of those three costs.

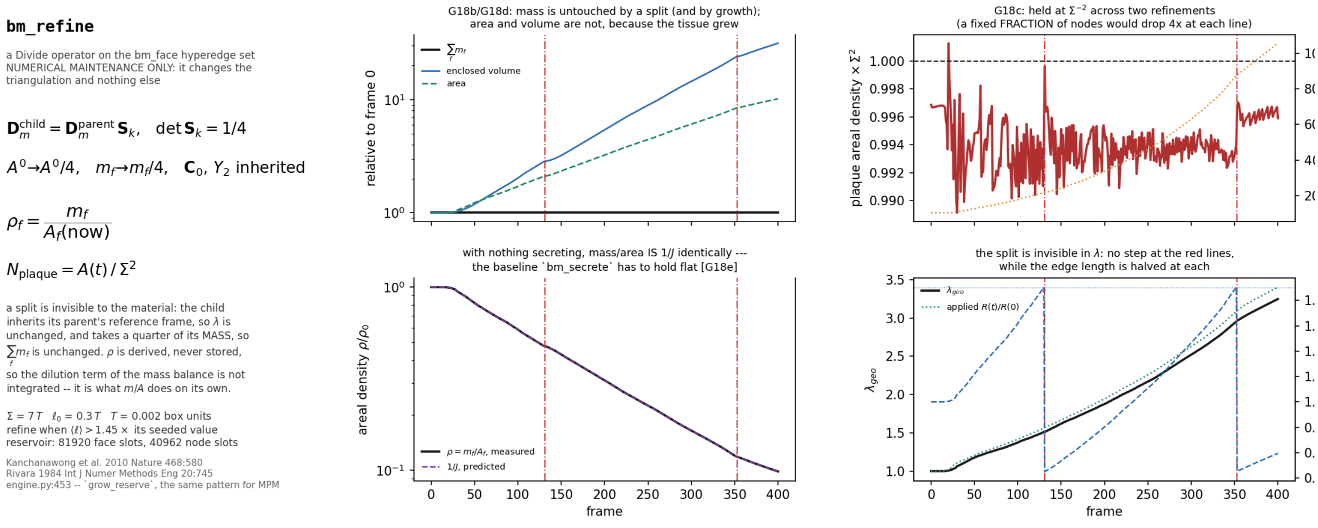

Figure 3: `bm_refine` and what it must carry (`05e_conserve`). Red dash-dot lines are the two refinements, at frames 131 and 353. **Top left G18b, G18d**: $\sum_f m_f$ is flat through both splits and through an area that grows ×11 — bit-identical — while the volume and area grow because the tissue did. **Top right G18c**: the plaque areal density is held at 0.996 of Σ^{-2}; the fixed-fraction rule 05b used ends at 0.145. **Bottom left** G18e: with nothing secreting, m/A IS $1/J$ identically — an identity, drawn because it is the baseline `bm_secrete` has to hold flat. **Bottom right** the split is invisible in λ (no step at the red lines) while the mean edge length is halved at each.

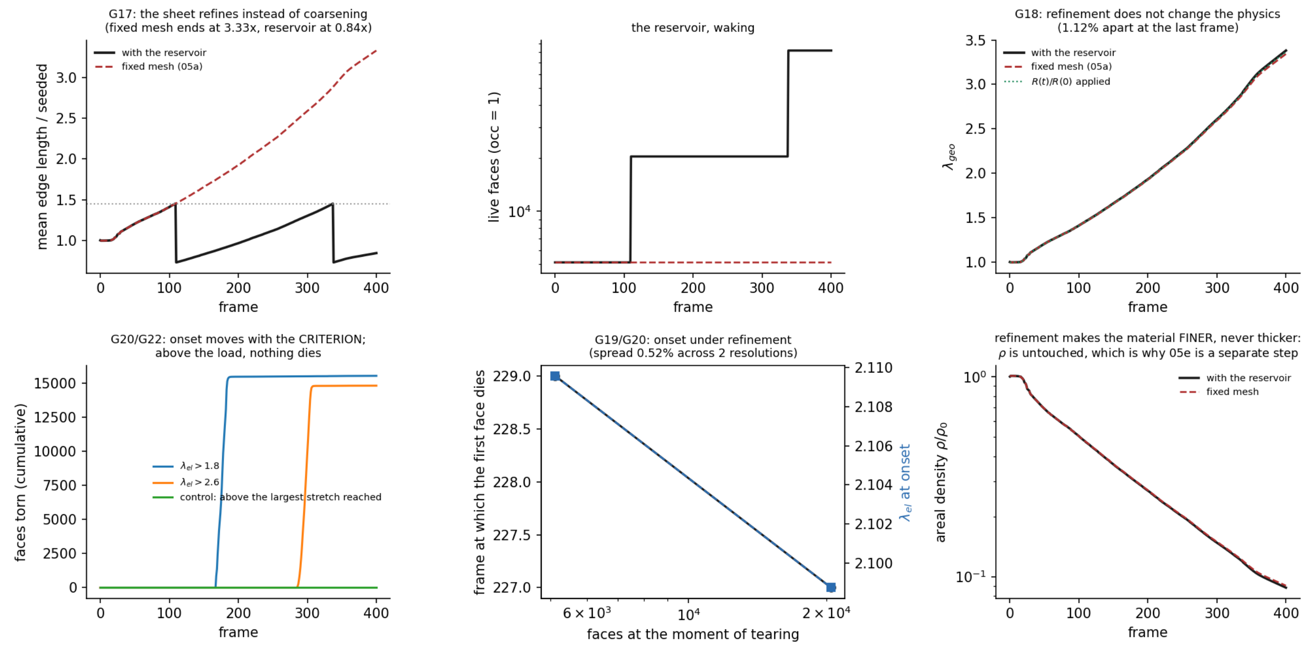

Figure 4: `05d_tear`: tearing, and the gate the MPM sheet was never given. **Bottom middle G19**: the same criterion at 5,115 and 20,471 faces tears at $\lambda^{\text{el}} = 2.110$ and 2.099 — 0.52% apart, against a 5% threshold — so the tear is set by the material and not by the discretisation. **Bottom left G20, G22**: onset moves with the criterion (frames 167 / 227 / 285 at thresholds 1.8 / 2.2 / 2.6) and a criterion above the largest stretch reached kills 0 of 81,920 faces — which is the run-127 control, run on purpose. **Top row G17, G18**: the reservoir holds the edge length at 0.84 of seeded where a fixed mesh reaches 3.33, and refinement improves the fidelity rather than merely preserving it.

4.1 The sweeps behind the table

`05a_sheet` — the sheet alone, tethered to the recorded epithelial surface (`smap`: 32 × 64 directions × 402 frames, R going 0.0398 → 0.1351 box units, a ratio of 3.387), 2562 nodes, 5120 faces, no matrix and no plaque.

run	τ_r	λ^{geo}	λ^{el}	standoff	what it says
nominal	∞	3.344	3.344	-1.52×10^{-3}	the sheet stores all of it
`tau100`	100 fr	3.384	1.226	-9.96×10^{-6}	153× better
`tau25`	25 fr	3.384	1.050	-1.48×10^{-6}	**1026**× better
`beta5`	∞	3.063	3.063	-1.29×10^{-2}	stiffening makes it worse
control $\kappa = 0$	—	1.0000	1.0000	—	the measure invents nothing
control rigid	—	1.0000	1.0000	—	motion is not deformation

The third row is a mechanism, not a tuning. Without remodelling the sheet's own hoop stress pulls it inward faster than a stable tether can hold it out — which is what `HANDOVER.md` concluded from runs 149–155 and could not test, because `membrane_tau` was 0 in every one of them. Turning it on moves the standoff by three orders of magnitude *while* λ^{geo} *stays at* 3.38: the sheet still goes where the tissue puts it, it simply stops fighting. $\beta = 5$ is the same statement from the other side — a sheet that stiffens as it stretches fights harder, and its standoff is 8.5× worse at $E(\lambda)/E_0 = 11.3$.

`05b_plaque` — 2562 membrane nodes against a 642-vertex epithelium, 2562 plaques, $\ell_0 = 0.3T$. G9 and G10 pass at machine precision, which is what this run exists for: the reaction is present, and present as motion and not only as bookkeeping. The density sweep, against the archived one-sided tether:

anchored	spacing	standoff, two-sided plaque	standoff, one-sided tether (archived)
100%	$1.5\,T$	$+1.46 \times 10^{-3}$	-8.2×10^{-3}
20%	$3.4\,T$	$+6.19 \times 10^{-3}$	-4.5×10^{-2}
4.6%	$7\,T$ (the measured Σ)	$+1.68 \times 10^{-2}$	-1.19×10^{-1}

At the density the literature measures — plaques $\sim$350 nm across on $\sim$700 nm centres, which on this mesh is 4.6% of the nodes — the two-sided plaque sits 7× closer to the surface than the one-sided tether did. The sign has flipped, and that is §10.

`05c_remesh` — the same sheet with two levels of reserve (81,920 face slots, 40,962 node slots), against the identical run with the reserve switched off. Refinement fires twice, at frames 110 and 338, each time when the mean edge reaches 1.45× its seeded value: 5120 → 20480 → 81920 faces, 2562 → 10242 → 40962 nodes.

	faces	mean edge / seeded	λ^{geo}	fidelity vs applied 3.387	standoff
reservoir, 2 levels	81,920	**0.84**	3.3817	**0.9985**	-9.89×10^{-5}
fixed mesh (05a)	5120	3.33	3.3440	0.9873	-1.52×10^{-3}

G17 passes. G18 was written as "the refined run must match the fixed-mesh run to 1%" and the two are 1.12% apart — **which is the gate being wrong, not the run.** The fixed mesh is not the truth: a coarse triangle chords a curved, stretching surface and under-reports how much it stretched, so refinement moves the fidelity from 0.9873 to 0.9985 and the residual error from 1.3% to 0.15%. The gate is therefore restated against the *applied* stretch, where refinement can only help, and G18 as originally phrased would have rejected an improvement. The standoff improves by 15× in the same run, for the same reason: more nodes tethered to the surface track it more closely.

`05e_conserve` — the reservoir and the plaque in one rig, so that a refinement has something to break, with mass as the state (Eq. 9) and `plaque_seed` holding Σ^{-2}. Over 401 frames it refines twice, at frames 131 and 353, and the sheet's area goes ×11:

	threshold	measured	
G18b	$\sum_f m_f$ across a split	exact	**0**, and **0** over the whole run
G18c	plaque density $\times\Sigma^2$	within 10%	**0.996** (104 → 1056 plaques)
G18c	*control*: 05b's fixed-fraction rule	—	0.145
G18d	volume / area / centroid across a split	$< 10^{-12}$	2.6×10^{-16}, **0**, 9.1×10^{-17}
G18e	$\rho = m/A$ against $1/J$	—	1.4×10^{-16}: an identity, not a test

Two things in that table are worth saying in words. **Mass is bit-identical across a split and across the whole run**, which is what licenses `bm_secrete` to be built on top: a substrate that lost material when it refined could not tell secretion from arithmetic. And **the control is a larger failure than predicted**: 05b's rule loses a factor of 6.9, not the factor of 4 per refinement one would guess, because a fixed fraction is wrong at each split *and* continuously as the area grows between them.

One operator changed kind because of this. `bm_secrete` was written as a *Divide* — wake reserve faces to add material, which is what a particle model must do. With mass carried as a scalar on a face it is a *Lateral* operator: it raises m_f on faces that already exist, and the reservoir is needed for resolution and not for supply. That is a simplification the ρ-as-state form would have hidden.

4.2 Three gates that failed, and none of them is a parameter

The standoff is $3.43\,\ell_0$, not ℓ_0 (G12). The sheet lags the expanding epithelium, so every plaque is *extended* rather than sitting at its rest length, and the extension is what carries the sheet outward. ℓ_0 therefore sets the standoff only in a quasi-static sheet; in a growing one it is ℓ_0 plus a lag, and the lag is the same hoop stress §9 isolates. The prediction is sharp and is the next run: `05b` with $\tau_r = 25$ should collapse $3.43 \to\sim 1$.

Friction is currently unmeasurable in this rig (G13). With $\xi = 0$ the sheet twisted 28.49° against a drive of 28.50° — it followed the epithelium essentially exactly with no friction at all. The reason is structural: a plaque anchored to a *fixed* barycentric point is a pin in the tangential direction, because that anchor is a material coordinate that rotates with the face. ξ can only set how fast the sheet equilibrates to a pin, never whether it slides. **Slip requires the anchor to move over the epithelial surface**, i.e. `plaque_seed` as turnover. §5 of `note_spheroid_bm_ecm` asked for friction where what was needed was turnover; that is a correction to the design, and the run is what found it.

Rupture is all-or-nothing for a structural reason. At a threshold of 0.0052, inside the measured load range (the nominal's p99 reaches 0.0120), *every* plaque broke. With a load that rises monotonically as the tissue grows and nothing that re-forms a bond, any threshold inside the range is reached by every plaque eventually: the fraction bound is a function of time and not of load. A partial-rupture regime needs a distribution of thresholds across plaques, or turnover. Reporting a fraction bound without one of those would be reporting a clock.

4.3 What this licenses, and what it does not

Licensed: the sheet's deformation is measured rather than advected (G1–G5); its stability limit is a rate that is measured rather than asserted (G8); its adhesion is a relation that returns its reaction to machine precision (G9, G10); and its *size* is now a state variable that can be changed without disturbing the mechanics at all (G14–G16, with area and energy bit-identical across a split). Those are the four things the archived line could not get, and they are what `04` needs before a membrane can be put into it.

Not licensed: the standoff is not yet ℓ_0; slip needs turnover, not friction; rupture needs a threshold distribution; there is no secretion, so Eq. (8) is a baseline ($\rho \to 1/11.1$) and not yet a mechanism; refinement is global rather than adaptive, and a crack tip will want local; there is no coarsening, so the sheet can only get finer; there is no bending stiffness, so it cannot fold, and a hole's rim has no line tension; there is no matrix and no `fibril`, so nothing yet removes the epithelium's direct push on the stroma; and the epithelium in `05b` is a driven icosphere, not the vertex model — `mesh_contact` certifies that interface separately, and joining the two is what `05g` does.

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